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DSC650 – Big Data
Professor Nasheb Ismaily
December 14, 2025

Week 2 Exercise

Objective 1. Conceptual Foundations

This week's exercise focuses on HDFS (Hadoop Distributed File System), YARN, and MapReduce, which are core components of the Hadoop ecosystem. Together, these tools form the foundation of Hadoop's distributed architecture and demonstrate how large-scale data can be stored, accessed, and processed using industry-standard practices. Hadoop is designed to handle petabytes of data efficiently by distributing both storage and computation across many machines, allowing organizations to scale while maintaining reliability and performance.

HDFS serves as the storage layer of the Hadoop architecture. Instead of storing data on a single system, HDFS breaks large files into blocks and distributes them across multiple DataNodes in the cluster. Each block is replicated across several nodes, providing built-in fault tolerance and ensuring data availability even if a node fails. This design allows Hadoop to store massive datasets reliably while also enabling faster data access by moving computation closer to where the data resides.

YARN, which stands for Yet Another Resource Negotiator, is responsible for resource management and job scheduling within the Hadoop cluster. Similar in concept to hypervisors or virtualization platforms like VMware, YARN allocates memory and CPU resources by managing containers and Application Masters across nodes. Because cluster resources are limited, YARN plays a critical role in prioritizing workloads, balancing resource usage, and ensuring that applications run efficiently and reliably without overwhelming the system.

Gabriel Vance
DSC650 – Big Data
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December 14, 2025

MapReduce is the original distributed processing model used by Hadoop to analyze large datasets in parallel. Data is divided into input splits, which are processed by Map tasks that generate intermediate key-value pairs. These results are then shuffled and sorted before being passed to Reduce tasks, which aggregate and produce the final output. This model demonstrates how large-scale computations can be performed efficiently by distributing work across multiple nodes in the cluster.

Objective 2. Deep Dive into HDFS

This week's exercise focuses on HDFS and its interaction with the Hadoop ecosystem. We began by running the `hdfs dfsadmin -report` command, which provides a detailed overview of the HDFS cluster's current state. This report includes information about the health of the cluster, available storage capacity, and the status of each DataNode. Reviewing this output is an important first step in understanding how data is

Gabriel Vance
DSC650 – Big Data
Professor Nasheb Ismaily
December 14, 2025
distributed and managed across the cluster.

```
-------------------------------------------------
Live datanodes (2):

Name: 172.28.1.2:9866 (worker1)
Hostname: worker1
Decommission Status : Normal
Configured Capacity: 50839412736 (47.35 GB)
DFS Used: 171170800 (163.24 MB)
Non DFS Used: 18430772240 (17.16 GB)
DFS Remaining: 31952259395 (29.76 GB)
DFS Used%: 0.34%
DFS Remaining%: 62.85%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 5
Last contact: Mon Dec 15 01:17:27 GMT 2025
Last Block Report: Mon Dec 15 01:13:06 GMT 2025
Num of Blocks: 140


Name: 172.28.1.3:9866 (worker2)
Hostname: worker2
Decommission Status : Normal
Configured Capacity: 50839412736 (47.35 GB)
DFS Used: 209033431 (199.35 MB)
Non DFS Used: 18392942377 (17.13 GB)
DFS Remaining: 31952252486 (29.76 GB)
DFS Used%: 0.41%
DFS Remaining%: 62.85%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 5
Last contact: Mon Dec 15 01:17:29 GMT 2025
Last Block Report: Mon Dec 15 01:13:06 GMT 2025
Num of Blocks: 136
```

To demonstrate HDFS storage, we uploaded grades.csv into the HDFS root directory using `hdfs dfs -put`. HDFS stores the file as one or more blocks on DataNodes and replicates those blocks based on the cluster's replication factor. We can verify which

Gabriel Vance
DSC650 – Big Data
Professor Nasheb Ismaily
December 14, 2025
DataNodes hold the blocks using HDFS tools (such as fsck), rather than expecting to find a normal ‘grades.csv’ file on each worker’s local filesystem.

```
bash-5.0# hdfs dfs -put /data/grades.csv /
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/li
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.1
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanat
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2025-12-15 01:19:18,728 WARN util.NativeCodeLoader: Unable to load native-ha
bash-5.0# hdfs dfs -ls /
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/li
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.1
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanat
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2025-12-15 01:19:55,119 WARN util.NativeCodeLoader: Unable to load native-ha
Found 7 items
-rw-r--r--   1 root supergroup        747 2025-12-15 01:19 /grades.csv
drwxr-xr-x   - root supergroup          0 2025-12-15 01:16 /hbase
drwxr-xr-x   - root supergroup          0 2025-12-15 01:14 /log
drwxr-xr-x   - root supergroup          0 2025-12-15 01:16 /spark-jars
drwxr-xr-x   - root supergroup          0 2025-12-15 01:16 /tez
drwxr-xr-x   - root supergroup          0 2025-12-15 01:14 /tmp
drwxrwx---   - root supergroup          0 2025-12-15 01:15 /user
bash-5.0#
```

`docker compose exec worker1/worker2 bash` opens a shell inside each worker container, but `hdfs dfs -ls /` always shows the same output because it lists the shared HDFS namespace managed by the NameNode, not the local filesystem. This confirms the file exists in HDFS and is visible cluster-wide, though it does not indicate which workers physically store the file’s data blocks.

Gabriel Vance
DSC650 – Big Data
Professor Nasheb Ismaily
December 14, 2025

```
bash-5.0# hdfs dfs -ls /
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/li
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.1
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanat
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2025-12-15 01:23:04,122 WARN util.NativeCodeLoader: Unable to load native-ha
Found 7 items
-rw-r--r--   1 root supergroup        747 2025-12-15 01:19 /grades.csv
drwxr-xr-x   - root supergroup          0 2025-12-15 01:16 /hbase
drwxr-xr-x   - root supergroup          0 2025-12-15 01:14 /log
drwxr-xr-x   - root supergroup          0 2025-12-15 01:16 /spark-jars
drwxr-xr-x   - root supergroup          0 2025-12-15 01:16 /tez
drwxr-xr-x   - root supergroup          0 2025-12-15 01:14 /tmp
drwxrwx---   - root supergroup          0 2025-12-15 01:15 /user
bash-5.0#
```

```
gvance67@dsc-650-gvance:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker compose exec worker2 bash
WARN[0000] /home/gvance67/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase/docker-compose.yml: the attribute
emove it to avoid potential confusion
bash-5.0# hdfs dfs -ls /
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBi
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2025-12-15 01:26:39,803 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... usin
Found 7 items
-rw-r--r--   1 root supergroup        747 2025-12-15 01:19 /grades.csv
drwxr-xr-x   - root supergroup          0 2025-12-15 01:16 /hbase
drwxr-xr-x   - root supergroup          0 2025-12-15 01:14 /log
drwxr-xr-x   - root supergroup          0 2025-12-15 01:16 /spark-jars
drwxr-xr-x   - root supergroup          0 2025-12-15 01:16 /tez
drwxr-xr-x   - root supergroup          0 2025-12-15 01:14 /tmp
drwxrwx---   - root supergroup          0 2025-12-15 01:15 /user
bash-5.0#
```

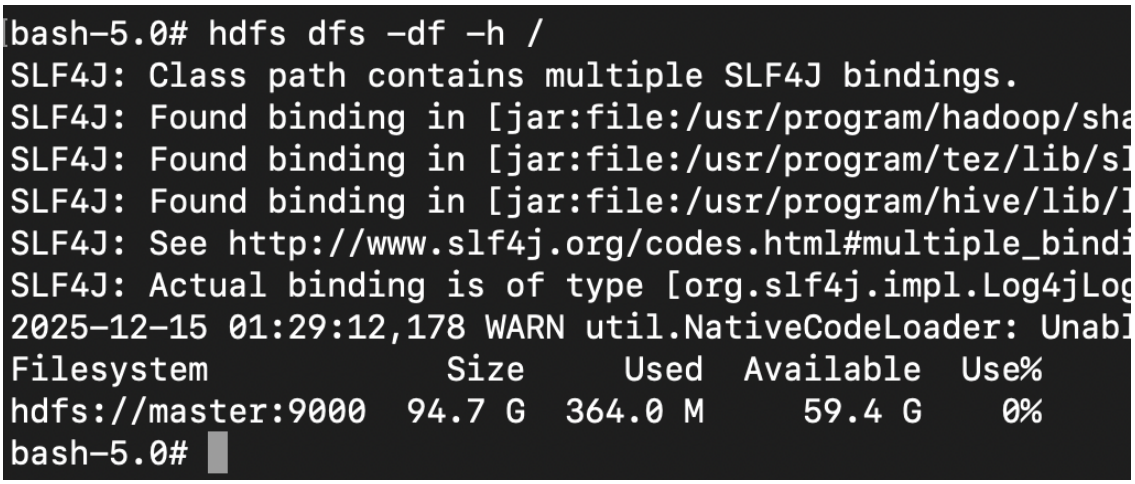
The `df -h` command reports disk usage and available capacity in a human-readable format, allowing us to understand how much storage space is used and remaining across the filesystem.

The `fsck` command performs a filesystem check on HDFS and reports detailed information about file health, including block locations, replication status, and whether any blocks are missing or corrupted.

The `getfacl` command displays the access control lists for files and directories, showing detailed permission rules such as read, write, and execute access for users and groups.

Gabriel Vance
DSC650 – Big Data
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Command 1 - `df -h`



Command 2 - `fsck`

Gabriel Vance
DSC650 – Big Data
Professor Nasheb Ismaily
December 14, 2025

```
bash-5.0# hdfs fsck /
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/sha
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/sl
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/l
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindi
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLog
2025-12-15 01:33:17,986 WARN util.NativeCodeLoader: Unabl
Connecting to namenode via http://master:9870/fsck?ugi=rc
FSCK started by root (auth:SIMPLE) from /172.28.1.1 for p

..
Status: HEALTHY
 Number of data-nodes:  2
 Number of racks:               1
 Total dirs:                    71
 Total symlinks:                0

Replicated Blocks:
 Total size:    377267317 B
 Total files:   285 (Files currently being written: 5)
 Total blocks (validated):      282 (avg. block size 1337
 Minimally replicated blocks:   282 (100.0 %)
 Over-replicated blocks:        0 (0.0 %)
 Under-replicated blocks:       0 (0.0 %)
 Mis-replicated blocks:         0 (0.0 %)
 Default replication factor:    1
 Average block replication:     1.0
 Missing blocks:                0
 Corrupt blocks:                0
 Missing replicas:              0 (0.0 %)

Erasure Coded Block Groups:
 Total size:    0 B
 Total files:   0
 Total block groups (validated):        0
 Minimally erasure-coded block groups:  0
 Over-erasure-coded block groups:       0
 Under-erasure-coded block groups:      0
 Unsatisfactory placement block groups: 0
 Average block group size:      0.0
 Missing block groups:          0
 Corrupt block groups:          0
```

Command 3 – `getfacl`

Gabriel Vance
DSC650 – Big Data
Professor Nasheb Ismaily
December 14, 2025

```
bash-5.0# hdfs dfs -getfacl /
SLF4J: Class path contains mult
SLF4J: Found binding in [jar:fi
SLF4J: Found binding in [jar:fi
SLF4J: Found binding in [jar:fi
SLF4J: See http://www.slf4j.org
SLF4J: Actual binding is of typ
2025-12-15 01:45:23,958 WARN ut
# file: /
# owner: root
# group: supergroup
user::rwx
group::r-x
other::r-x

bash-5.0#
```

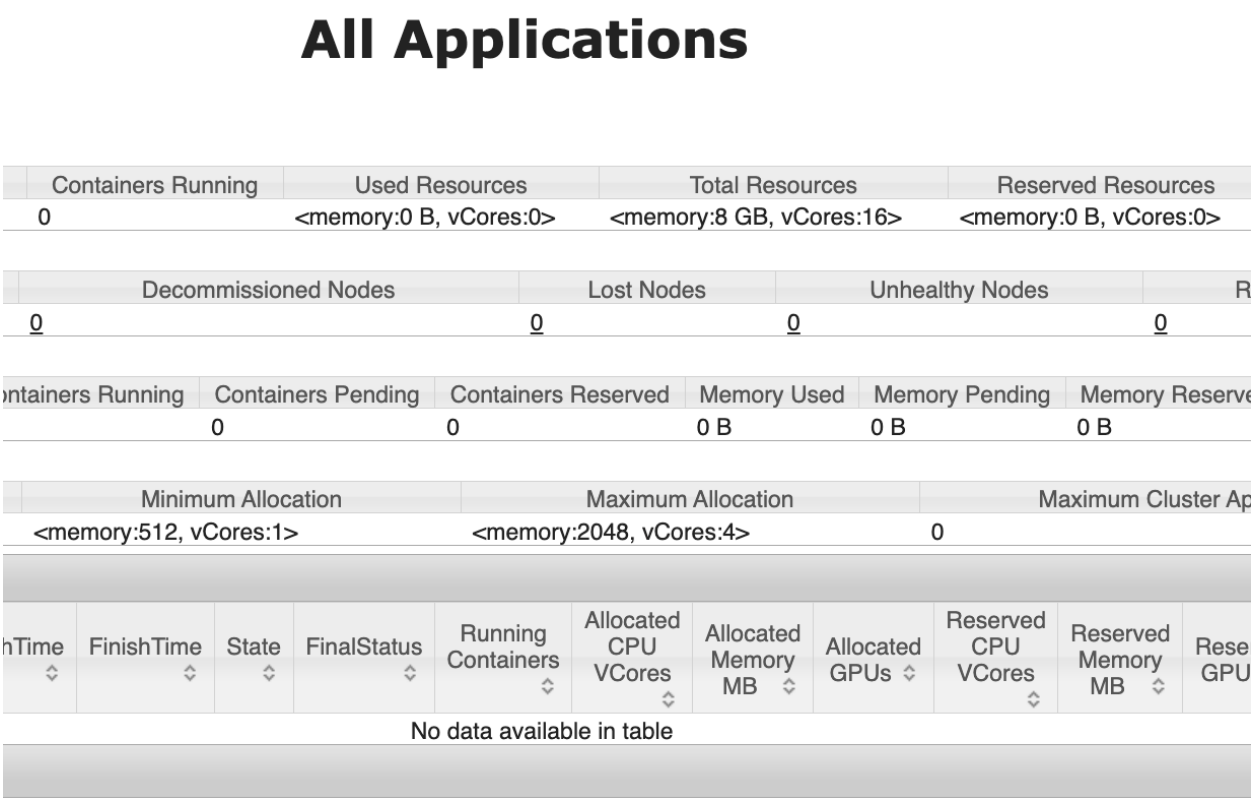
Objective 3 – Introduction to YARN

This command lists all YARN NodeManagers in the cluster and shows that two worker nodes are currently registered and running. The output also indicates that no containers are actively running on either node at the time of execution, meaning no YARN jobs are consuming resources.

```
bash-5.0# yarn node -list -all 2>/dev/null
Total Nodes:2
         Node-Id             Node-State Node-Http-Address       Number-of-Running-Containers
   worker1:37325                RUNNING      worker1:8042                                  0
   worker2:37981                RUNNING      worker2:8042                                  0
bash-5.0#
```

Gabriel Vance
DSC650 – Big Data
Professor Nasheb Ismaily
December 14, 2025

Within the `etc/hadoop` configuration directory, we have access to several Hadoop configuration files that control cluster behavior. Since the YARN ResourceManager application is running on port 8088, we can modify the memory-related settings to increase allocations to 2048 MB, and after applying the changes and refreshing the web interface, the updated memory values are reflected.



Objective 4. MapReduce Pi Job

The MapReduce Pi example uses a Monte Carlo simulation to estimate the value of π by randomly generating points within a square and counting how many fall inside a unit circle. Each map task independently performs this random sampling and reports its results, while the reduce task aggregates the counts across all maps to calculate the final approximation of π. This

Gabriel Vance
DSC650 – Big Data
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December 14, 2025
example illustrates how Hadoop can distribute probabilistic computations across multiple nodes

and combine the results efficiently using MapReduce.

```
bash-5.0# hadoop jar /usr/program/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.2.3.jar pi -libjars ${libjars} 2 10
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2025-12-15 02:22:58,286 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Number of Maps  = 2
Samples per Map = 10
Wrote input for Map #0
Wrote input for Map #1
Starting Job
2025-12-15 02:23:00,006 INFO client.RMProxy: Connecting to ResourceManager at master/172.28.1.1:8032
2025-12-15 02:23:00,472 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1765765288817_0001
2025-12-15 02:23:09,893 INFO input.FileInputFormat: Total input files to process : 2
2025-12-15 02:23:10,391 INFO mapreduce.JobSubmitter: number of splits:2
2025-12-15 02:23:11,393 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1765765288817_0001
2025-12-15 02:23:11,397 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-12-15 02:23:12,039 INFO conf.Configuration: resource-types.xml not found
2025-12-15 02:23:12,041 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-12-15 02:23:12,060 WARN mapred.YARNRunner: Configuration yarn.app.mapreduce.am.resource.memory-mb=512Mi is overriding the yarn.app.mapreduce.am.resource.mb=1536 configuration
2025-12-15 02:23:12,929 INFO impl.YarnClientImpl: Submitted application application_1765765288817_0001
2025-12-15 02:23:13,065 INFO mapreduce.Job: The url to track the job: http://master:8088/proxy/application_1765765288817_0001/
2025-12-15 02:23:13,067 INFO mapreduce.Job: Running job: job_1765765288817_0001
2025-12-15 02:23:29,345 INFO mapreduce.Job: Job job_1765765288817_0001 running in uber mode : false
2025-12-15 02:23:29,347 INFO mapreduce.Job:  map 0% reduce 0%
2025-12-15 02:23:35,457 INFO mapreduce.Job:  map 50% reduce 0%
2025-12-15 02:23:39,510 INFO mapreduce.Job:  map 100% reduce 0%
2025-12-15 02:23:42,541 INFO mapreduce.Job:  map 100% reduce 100%
2025-12-15 02:23:42,553 INFO mapreduce.Job: Job job_1765765288817_0001 completed successfully
2025-12-15 02:23:42,725 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=50
                FILE: Number of bytes written=738258
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=522
                HDFS: Number of bytes written=215
                HDFS: Number of read operations=13
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=3
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=2
```

Gabriel Vance
DSC650 – Big Data
Professor Nasheb Ismaily
December 14, 2025

```
        Job Counters
                Launched map tasks=2
                Launched reduce tasks=1
                Data-local map tasks=2
                Total time spent by all maps in occupied slots (ms)=23858
                Total time spent by all reduces in occupied slots (ms)=15852
                Total time spent by all map tasks (ms)=11929
                Total time spent by all reduce tasks (ms)=3963
                Total vcore-milliseconds taken by all map tasks=11929
                Total vcore-milliseconds taken by all reduce tasks=3963
                Total megabyte-milliseconds taken by all map tasks=12215296
                Total megabyte-milliseconds taken by all reduce tasks=8116224
        Map-Reduce Framework
                Map input records=2
                Map output records=4
                Map output bytes=36
                Map output materialized bytes=56
                Input split bytes=286
                Combine input records=0
                Combine output records=0
                Reduce input groups=2
                Reduce shuffle bytes=56
                Reduce input records=4
                Reduce output records=0
                Spilled Records=8
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=250
                CPU time spent (ms)=2190
                Physical memory (bytes) snapshot=762437632
                Virtual memory (bytes) snapshot=7068585984
                Total committed heap usage (bytes)=699924480
                Peak Map Physical memory (bytes)=286937088
                Peak Map Virtual memory (bytes)=2355445760
                Peak Reduce Physical memory (bytes)=189005824
                Peak Reduce Virtual memory (bytes)=2361114624
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=236
        File Output Format Counters
                Bytes Written=97
Job Finished in 42.931 seconds
Estimated value of Pi is 3.80000000000000000000
bash-5.0#
```